AI Profile Generator

Prompt Engineering Research Report

Project: CheckinMe

Model: Google Gemini 2.5 Flash Image (configurable)

Scope: Prompt design, comparative benchmarking & tooling

Date: 08 May 2026

Status: Experimental — benchmark runs in progress

This report documents a structured prompt engineering initiative to improve face identity preservation, skin tone fidelity, image sharpness, and background uniformity in AI-generated professional headshots. Four prompt versions are designed, benchmarked, and compared against a seven-metric quantitative framework.

Table of Contents

1. Executive Summary

CheckinMe's AI Profile Generator uses Google Gemini to transform a user-supplied reference photograph into a set of professional headshots with varied attire. The primary quality requirement — and the most common failure mode — is **face identity preservation**: the generated image must be recognisably the same person as in the input photo.

This report documents a structured prompt engineering effort to improve generation quality across four measurable dimensions: face structural similarity, skin tone fidelity, image sharpness, and background uniformity. Four prompt versions were designed (one production baseline and three experimental variants), and a Python/Streamlit benchmarking application was built to evaluate them against a seven-metric quantitative framework augmented by human qualitative assessment.

Key Contributions

- Identification of three structural weaknesses in the production prompt.
- Design of three experimental prompt variants, each addressing a different hypothesis about model instruction following.
- A reproducible benchmarking tool supporting side-by-side visual comparison, automated metric computation, and human rating capture.
- A configurable model selector enabling cross-model comparisons without code changes.
- A live prompt editor allowing real-time prompt modifications and A/B testing directly in the Streamlit UI.

2. Project Context

2.1 System Overview

The production generation pipeline is implemented in `AiProfileGenerateService.php` (Laravel). When a user requests headshots, the service:

- Accepts a reference image path, gender, quantity (limit), and rotation offset.
- Selects 'limit' outfit descriptions from a pool of 16 per gender, applying the offset as a circular rotation.
- Constructs one text prompt per outfit.
- Calls the Gemini multimodal endpoint (gemini-2.5-flash-image:generateContent) with the prompt and the base64-encoded reference image.
- Extracts the inlineData.data field from the API response and returns it as a base64 PNG.

In non-production environments the service short-circuits and returns random existing media URLs to avoid API cost during development.

2.2 Outfit Catalogue

Gender	Pool Size	Style Range
Male	16	Black/navy/charcoal/grey suits with varied shirt and tie combinations
Female	16	Black/navy/grey/dark blazers over white and light-coloured blouses

All outfits are conservative corporate styles deliberately selected to minimise the risk of anatomical distortion at the neckline, a known failure mode of AI portrait generators.

3. Problem Statement

3.1 Primary Failure Mode: Identity Drift

When a generative vision model is instructed to re-dress a subject, it faces a fundamental tension: it must synthesise new clothing while holding the face constant. Without sufficiently strong identity anchoring instructions, models tend to:

- Smooth or 'beautify' skin texture, altering the perceived person.
- Shift skin tone slightly warm or cool depending on the chosen outfit's colour temperature.
- Subtly reshape the nose, jaw, or eye area to fit a 'professional' archetype.
- Alter hair style or colour.

The combined effect is a headshot that looks professional but is not recognisably the same individual — a product failure.

3.2 Secondary Failure Modes

Failure Mode	Visible Symptom	Affected Metric
Skin tone drift	Complexion appears lighter/darker or different undertone	Skin Tone Match (20%)
Over-smoothing	Skin looks plastic or AI-generated	Face SSIM (25%), Face Sharpness (20%)
Framing errors	Head cropped, or too much torso shown	Framing Quality (10%)
Background noise	Gradient, texture, or vignette at corners	Background Uniformity (15%)
Anatomical distortion	Floating collar, misaligned neck	Qualitative — Clothing Quality

3.3 Structural Weaknesses in the Production Prompt

Analysis of `build_production_prompt()` reveals three structural issues:

Issue 1 — Late identity anchoring

The production prompt opens with outfit instruction (*Generate a portrait of a professional [gender] [outfit]*) before stating face preservation requirements. This ordering implicitly signals that the primary task is outfit generation, with identity preservation as a constraint. Multimodal LLMs tend to weight earlier instructions more heavily during synthesis.

Issue 2 — Absent skin tone instruction

The phrase 'Do not alter facial structure, skin texture, or expression' does not mention skin colour or undertone explicitly. Skin tone shift is measured at 20% of the quantitative score, making this a high-impact omission.

Issue 3 — Ambiguous background specification

'A soft, solid, professional ID card style background' is semantically underspecified. The benchmark measures background uniformity by sampling pixel variance in the four image corners. Any gradient or subtle texture that extends to the corners penalises the score; the production prompt does not instruct the model to maintain consistency to the image edges.

4. Technical Architecture

4.1 Production Service (AiProfileGenerateService.php)

```
User request
    ■
    ■■■ getPrompts(gender, limit, offset, aspectRatio)
    ■       ■■■ Selects outfits from $outfits[gender]
    ■       ■■■ Applies circular offset rotation
    ■       ■■■ Assembles: $commonIntro + $style + $commonFeatures
    ■                      + $framingFix + $neckFix + $imageQuality
    ■
    ■■■ foreach $prompts as $prompt
            ■■■ POST /v1beta/models/gemini-2.5-flash-image:generateContent
                    ■■■ parts[0]: text prompt
                    ■■■ parts[1]: inline_data (base64 PNG/JPEG)
                    → returns base64 image → stored / returned to controller
```

4.2 Benchmarking Application (benchmark_app.py)

```
Streamlit UI
    ■
    ■■■ Sidebar
    ■       ■■■ Gemini API Key
    ■       ■■■ Model selector (dynamic URL construction)
    ■       ■■■ Gender / Outfit builder / Background selector
    ■       ■■■ Experimental prompt version selector (v1 / v2 / v3)
    ■
    ■■■ Prompt Editor (main area)
    ■       ■■■ Production text area — editable, reset button, live char count
    ■       ■■■ Experimental text area — editable, reset button, live char count
    ■               Auto-reloads when version/outfit/background changes (fingerprint)
    ■
    ■■■ Generate & Compare (button)
    ■       ■■■ ThreadPoolExecutor(max_workers=2)
    ■               ■■■ call_gemini_api(prod_prompt, gemini_url)
    ■               ■■■ call_gemini_api(exp_prompt, gemini_url)
    ■
    ■■■ Side-by-Side Image Comparison
    ■
```

```
    ■■■ Quantitative Metrics  ■  bm.compute_all() → 8 scores
    ■
    ■■■ Qualitative Rating Panel (5 criteria × 1–5 stars)
            ■■■ Final = 60% Quantitative + 40% Qualitative
```

4.3 Model URL Architecture

The Gemini model endpoint follows the pattern:

```
https://generativelanguage.googleapis.com/v1beta/models/{MODEL_ID}:generateContent
```

The model ID was previously hardcoded as the module-level constant `GEMINI_URL`. It is now constructed dynamically from a sidebar selectbox, enabling cross-model comparison (e.g., `gemini-2.5-flash-image` vs `gemini-2.0-flash-exp-image-generation`) without modifying source code. A custom model ID text input is also available for unreleased or preview models.

5. Evaluation Methodology

5.1 Quantitative Metrics

All metrics produce scores on a 0–100 scale, combined into an **Overall Quantitative Score** via a fixed weighted average.

Metric	Weight	Library	Primary Failure Detected
Face SSIM	25%	scikit-image	Identity drift — structural (SSIM on 96×96 face crop)
Skin Tone Match	20%	numpy	Skin colour / undertone shift (R+G histogram correlation)
Face Sharpness	20%	opencv	Blur, over-smoothing (Laplacian variance, threshold 1500)
Background Uniformity	15%	numpy	Gradients, textures, vignettes (corner std dev)
Framing Quality	10%	opencv	Zoom too close or too far (face/image height ratio)
Face Centering	5%	opencv	Off-centre composition (horizontal centroid offset)
Exposure Quality	5%	numpy	Over/underexposure (mean luminance vs 90–175 range)

> **Overall Quantitative Score = SSIM×0.25 + Skin×0.20 + Sharp×0.20 + BG×0.15 + Framing×0.10 + Centering×0.05 + Exposure×0.05**

5.2 Face Detection

Pre-processing for SSIM, Skin Tone, Sharpness, Framing, and Centering depends on locating the face. The implementation uses OpenCV's Haar cascade (`haarcascade_frontalface_default.xml`) with `scaleFactor=1.1`, `minNeighbors=5`, `minSize=(40,40)`. If detection fails, affected metrics return a neutral score of 50 (or 0 for Framing/Centering).

5.3 Qualitative Assessment

Criterion	Question
Face Resemblance	Does the face look exactly like the person in the reference photo?
Professional Look	Does the overall image look like a professional headshot?
Clothing Quality	Does the clothing look natural, well-fitted, and realistic?
Background Quality	Is the background clean, solid, and professional?
Overall Preference	Overall, how satisfied are you with this image?

```
Qualitative Score = (mean star rating / 5) × 100
```

5.4 Final Combined Score

```
Final = 0.60 × Quantitative_Overall + 0.40 × Qualitative_Average
```

Condition	Verdict
$|Exp – Prod| < 2$	Tie
Exp – Prod ≥ 2	Experimental wins
Prod – Exp ≥ 2	Production wins

6. Prompt Engineering Analysis

6.1 Design Philosophy

Each experimental version tests a distinct hypothesis about how Gemini processes multimodal instructions:

Version	Hypothesis
Production	Baseline: implicit identity constraint appended after outfit instruction.
v1 — Photographer Role	Assigning a domain expert persona improves instruction compliance for professional outputs.
v2 — Step-by-Step Identity	Sequential processing steps with explicit metric-aligned sub-instructions improve targeted output quality.
v3 — Directive + Prohibitions	Explicit negative constraints (✗ DO NOT) are more reliably honoured than equivalent positive instructions.

6.2 Production Prompt (Baseline)

Attribute	Value
Approximate length	~430 characters / ~75 words
Instruction order	Outfit → Identity → Framing → Anatomy → Quality
Skin tone mentioned	No
Hair preservation	No
Background edge spec	No
Negative prohibitions	No

Strengths: Concise, low token overhead, includes neck/anatomy fix and framing instruction.

Weaknesses: Outfit is the first clause — identity anchoring is secondary. No explicit skin tone, hair, age, or ethnic feature preservation. Background described as 'ID card style' — semantically ambiguous for corner uniformity. All instructions are positive; no negation list.

6.3 Experimental v1 — Photographer Role

Attribute	Value
Approximate length	~900 characters / ~160 words
Role assignment	You are a professional portrait photographer
Identity before outfit	Yes
Skin tone named	Partially (within identity bullet)

Attribute	Value
Hair preservation	Yes
Negative prohibitions	No

Key structural change: Assigns a domain expert role before any task specification. Role-prompting in LLM literature shifts model behaviour toward domain-appropriate defaults. Identity block now precedes outfit instruction.

Remaining gaps: Skin tone and undertone not called out by name. Background uniformity to image edges not explicitly required. No negative prohibition block.

6.4 Experimental v2 — Step-by-Step Identity

Attribute	Value
Approximate length	~1,150 characters / ~200 words
Structure	6 numbered processing steps
Skin tone named	Yes — Step 1 first bullet, incl. undertone
Background edge spec	Yes — 'clean all the way to the image edges'
Sharpness instruction	Yes — 'face razor-sharp in focus'
Negative prohibitions	No

Key structural change: Replaces free-form instruction blocks with numbered processing steps, explicitly sequencing the model to anchor identity first before applying styling.

Metric	Improvement Mechanism
Skin Tone Match	First bullet in Step 1 names skin tone and undertone explicitly; model processes identity before outfit colour.
Face SSIM	Identity anchoring in Step 1 before styling reduces structural drift.
Background Uniformity	Step 5 requires clean background 'all the way to the image edges'.
Face Sharpness	Step 6 instructs 'face razor-sharp in focus'.
Framing	Step 3 adds '5–10% headroom above the head' for reliable crown inclusion.

6.5 Experimental v3 — Directive + Prohibitions

Attribute	Value
Approximate length	~1,400 characters / ~240 words
Structure	5 headed sections + prohibitions block

Attribute	Value
Lighting specification	3-point studio (key + fill + rim)
Skin tone named	Yes — first identity bullet
Negative prohibitions	Yes — 6 explicit ✗ DO NOT rules
Background edge spec	Yes — 'consistent from center to all four edges'

Key structural changes: Abandons prose/bullet style for a structured brief with section dividers and a terminal STRICT PROHIBITIONS block. Tests whether explicit negative constraints (✗ DO NOT) are more reliably honoured than equivalent positive instructions. Adds a full 3-point studio lighting specification unique to this version.

Metric	Improvement Mechanism
Skin Tone Match	First identity bullet; skin tone both positively asserted and negatively prohibited.
Face SSIM	'Zero tolerance' framing + dual positive/negative constraints on facial geometry.
Exposure Quality	Explicit 3-point lighting with key/fill/rim spec; prohibition on HDR and cinematic grading.
Background Uniformity	'Consistent from center to all four edges' targets corner-sampling metric.

6.6 Prompt Comparison Summary

Dimension	Production	v1	v2	v3
Identity before outfit	✗	✓	✓	✓
Skin tone named explicitly	✗	✗	✓	✓
Hair preservation	✗	✓	✓	✓
Age preservation	✗	✗	✓	✓
Background edge uniformity	✗	✗	✓	✓
Sharpness / focus instruction	✗	✓	✓	✓
3-point lighting spec	✗	✗	✗	✓
Negative prohibition block	✗	✗	✗	✓
Approx. word count	~75	~160	~200	~240

7. Benchmarking Tool Design

7.1 Live Prompt Editor

The original application displayed prompts in a read-only expander. This was replaced with a live prompt editor — two side-by-side editable text areas with the following behaviour:

- **Auto-reload on fingerprint change:** A fingerprint string encodes gender | custom_outfit | experimental_outfit | background_style | exp_version_key. When any sidebar selection changes, both text areas reload with the updated preset.
- **Manual override:** Within a constant fingerprint, user edits persist across Streamlit re-runs. Exactly the edited text is sent to the API.
- **Reset buttons:** ■ Reset restores the text area to the current preset without changing other state.
- **Live statistics:** Character and word count displayed beneath each editor.

7.2 Experimental Version Selector

A radio widget in the sidebar presents the three experimental versions. Selecting a version changes the fingerprint, triggers auto-reload of the experimental text area, and labels the result image header with the version name for unambiguous identification.

7.3 Configurable Model Selector

The Gemini model is now constructed dynamically from a sidebar selectbox. `call_gemini_api` accepts `url` as an explicit parameter (defaulting to `GEMINI_URL` for backwards compatibility). Both production and experimental calls use the same model, ensuring fair comparison.

7.4 Parallel Generation

Both prompts are submitted concurrently using `ThreadPoolExecutor(max_workers=2)`, minimising total wall-clock time and ensuring that any temporal variation in the Gemini API (temperature, model load) affects both prompts equally.

8. Results

This section will be populated after benchmark runs are completed. The framework is ready for data collection.

8.1 Data Collection Protocol

For statistically meaningful results, each prompt version should be tested with:

- Minimum 3–5 different reference photos (diverse skin tones, genders, ages).
- At least 3 generations per photo per version, given Gemini's stochastic nature.
- Qualitative ratings completed immediately after each generation while the reference image is visible.

8.2 Expected Results (Hypothesis)

Metric	Expected Ranking
Face SSIM	v3 ≥ v2 > v1 > Production
Skin Tone Match	v3 ≥ v2 > v1 > Production
Face Sharpness	v2 ≈ v3 > v1 > Production
Background Uniformity	v2 ≈ v3 > v1 > Production
Framing Quality	v2 ≥ v1 ≈ v3 ≈ Production
Exposure Quality	v3 > v2 ≈ v1 ≈ Production
Overall Score	v3 ≥ v2 > v1 > Production

Note: Longer prompts are not always better. There is a risk that v3's structured format (Unicode dividers, ✗ symbols) is less well-handled by the model than plain prose, which could cause unexpected regressions. This hypothesis must be validated empirically.

8.3 Results Table (to be filled)

Run	Reference Photo	Gender	Version	Model	SSIM	Skin	Sharp	BG	Overall
—	—	—	—	—	—	—	—	—	—

9. Recommendations

9.1 Immediate

• Run the benchmark with 3+ reference photos before making any prompt changes to production. Never change a production prompt based on visual intuition alone.

• Use gemini-2.5-flash-image for all runs in the initial benchmark pass to ensure model is not a confounding variable.

• Rate qualitative scores immediately after generation — recency bias affects scoring accuracy if ratings are done in batch later.

9.2 Short-Term (pending benchmark results)

• If v2 or v3 outperforms production on Face SSIM + Skin Tone Match without regressing on Framing/Exposure, adopt the winner as the new production prompt in AiProfileGenerateService.php.

• Consider adding skin tone mention to the production prompt as a minimal low-risk improvement even before full experiment results are available, since it addresses a clear gap with no downside.

9.3 Medium-Term

• Replace Haar cascade with a deep-learning face detector (MTCNN or RetinaFace). Haar cascade fails on non-frontal faces and profiles.

• Add FaceNet / ArcFace cosine similarity as an 8th metric. SSIM measures structural patterns but is not identity-aware — embedding-based identity distance is a more principled measure.

• Average metrics across 3 generations per prompt to reduce per-run stochasticity. A single bad generation can skew Overall Score by 5–10 points.

• Add CLIP scoring ('professional business headshot' anchor text) as a semantic quality metric independent of the reference photo.

9.4 Long-Term

• Consider prompt personalisation by skin tone range: darker skin tones are more susceptible to brightening drift by the model.

• Evaluate Gemini 2.0 Flash Experimental as a model upgrade candidate using the configurable model selector added in this work.

10. Appendix — Full Prompt Texts

A Production Prompt (Baseline)

```
Generate a portrait of a professional {gender} {outfit}.
STRICTLY PRESERVE the subject's original face. The output must look exactly
like the person in the uploaded image. Do not alter facial structure, skin
texture, or expression. The body should be centered with a natural, professional
pose. Background must be a soft, solid, professional ID card style background.
Frame the image from the top of the head to the mid-chest. Ensure the neck and
shoulders are anatomically correct and proportional. The clothing must fit
naturally around the neck/collar area without floating or weird cuts.
PHOTOREALISTIC style. The image must look like a high-end photograph.
No cartoonish, 3D render, or filtered looks. Lighting should be natural studio
lighting.
```

B Experimental v1 — Photographer Role

```
You are a professional portrait photographer. Using the uploaded reference photo,
generate a photorealistic headshot portrait with the following specifications:

IDENTITY PRESERVATION (CRITICAL):
- The subject's face must be IDENTICAL to the person in the uploaded image —
  same skin tone, eye color, nose shape, jawline, and every facial feature
- Do NOT alter, enhance, smooth, or modify the face in any way
- Preserve exact skin texture, any facial marks, and natural expression
- Do not change hair color, hair style, or eyebrow shape

SUBJECT & COMPOSITION:
- Professional {gender} portrait
- {outfit}
- Framing: from the crown of the head to the mid-chest, face centered horizontally
- Natural, confident, upright posture — shoulders squared to camera

CLOTHING & ANATOMY:
- Clothing must be sharp, properly fitted, wrinkle-free
- Collar and neckline must sit naturally — no gaps, floating fabric, or distortion
- Neck and shoulders must be anatomically correct and proportional

```

```
BACKGROUND:
- {background}, clean professional background (LinkedIn / ID card portrait style)
- No props, no distractions — soft gradient or flat solid color only

PHOTOGRAPHY STYLE:
- Ultra-photorealistic — must look like a real photograph taken in a professional studio
- No artistic filters, no painterly effects, no 3D rendering
- Soft box studio lighting: main light slightly above-left, subtle fill from the right
- Face in sharp focus, background softly diffused (shallow depth of field)
- Color grading: neutral, true-to-life — no heavy warm/cool toning

OUTPUT: A single portrait image matching all specifications above.
```

C Experimental v2 — Step-by-Step Identity

```
You are processing a real reference photograph to generate a professional corporate
headshot. The face in the reference photo must appear EXACTLY in the output.

STEP 1 — ANCHOR THE IDENTITY (highest priority):
Study every detail of the face in the uploaded photo before generating:
  Skin tone and undertone (warm / cool / neutral) — match precisely
  Skin texture: pores, fine lines, any marks or freckles — do not smooth away
  Eye color, shape, and spacing
  Nose shape, lip shape, jawline, and cheekbone structure
  Hair: exact color, texture, and current style as shown
  Apparent age — do not de-age or age the person
  Ethnicity and all distinguishing features
This face must appear UNCHANGED in the output.

STEP 2 — APPLY PROFESSIONAL STYLING:
Subject: {gender} professional    Attire: {outfit}

STEP 3 — COMPOSITION:
  Framing: crown of head to mid-chest, 5-10% headroom above the head
  Face horizontally centered in frame
  Upright posture, shoulders relaxed, confident neutral expression

STEP 4 — CLOTHING & ANATOMY:
  Clothing crisp, well-tailored, wrinkle-free
  Collar sits flat against neck — no floating fabric, no gaps
```

```
  Neck and shoulder anatomy natural and proportional

STEP 5 — BACKGROUND:
  {background} — completely uniform solid or very subtle gradient
  Must be clean all the way to the image edges — no texture or environmental elements

STEP 6 — LIGHTING & FOCUS:
  Key light at 45° upper-left, gentle fill from right
  Even illumination — no harsh chin or nose shadows
  Face razor-sharp in focus; background may have slight depth-of-field softness
  Ultra-photorealistic — indistinguishable from a real studio photograph

OUTPUT: Single portrait image exactly matching the above specifications.
```

D Experimental v3 — Directive + Prohibitions

```
PROFESSIONAL HEADSHOT DIRECTIVE
INPUT: Reference photograph of a real person
OUTPUT: Professional {gender} corporate headshot

IDENTITY — ZERO TOLERANCE FOR CHANGES
The person in the generated image must be the EXACT SAME PERSON as in the
reference photo. Treat the face as a locked, uneditable asset:
  Skin color and undertone: match exactly — same warmth, depth, and tone
  Skin texture: preserve all natural texture, pores, fine lines — no AI smoothing
  Facial geometry: same bone structure, proportions, and all features
  Eyes: same color, shape, spacing, and brow arch
  Hair: same color and style as shown
  Age: no de-aging or aging — keep the subject's natural age

ATTIRE
{outfit}  |  Garment must be sharp, properly fitted, and wrinkle-free.
Collar and neckline sit naturally — no gaps or floating fabric.

FRAMING & POSE
  Head-to-mid-chest crop — full crown of head with small headroom margin
  Face centered horizontally
  Upright, professional posture — relaxed shoulders, no unnatural tilt

BACKGROUND
```

```
{background} — clean, uniform, professional studio backdrop
Completely free of texture, objects, or environmental context.
Background must be consistent from center to all four edges of the frame.

LIGHTING  (3-point studio setup)
  Key light — upper-left at 45°, soft box diffused
  Fill light — right side, softer intensity to open shadows
  Rim / hair light — subtle, from behind, to separate subject from background

TECHNICAL OUTPUT
  Photorealistic DSLR photograph — high resolution, no compression artifacts
  85mm portrait lens equivalent at f/2.2 — face tack-sharp, background soft
  No artistic filters, no painterly or HDR effects, no stylization

STRICT PROHIBITIONS
  Do NOT generate a different face
  Do NOT smooth, beautify, or retouch skin
  Do NOT change skin tone, hair color, or eye color
  Do NOT alter facial proportions
  Do NOT add accessories, props, or background elements not specified
  Do NOT apply cinematic, HDR, or stylized color grading
```